

ANJANEYULU PALLEPAGU

Mobile: +91-8463987840 | Email: anjaneyulupallepagu@gmail.com
LinkedIn: linkedin.com/in/anjaneyulu-pallepagu-bb103534b | Medium:
medium.com/@anjiepallepagu
Location: Bangalore, India

PROFESSIONAL SUMMARY

Highly analytical and impact-driven Senior Data & AI Engineer with over 4 years of enterprise-scale experience architecting robust data pipelines, end-to-end data models, and intelligent AI integration workflows. Proven track record managing over 50TB of daily data volume for Fortune 500 environments, notably leading a zero-downtime CDH-to-CDP enterprise migration that realized \$2M in annual infrastructure cost reductions. Expert in translating complex business requirements into high-performance structural data models, mapping entities, establishing relational integrity, and applying strong AI theory to build predictive systems and advanced automation solutions for top-tier global organizations.

TECHNICAL SKILLS

- * **Data Modeling & Design:** Conceptual, Logical, & Physical Data Modeling, Entity-Relationship Diagrams (ERD), Dimensional Modeling, Normalization, Data Lineage
- * **Big Data Technologies:** Apache Hadoop, PySpark, Hive, Impala, Sqoop, YARN, MapReduce, Apache Kafka.
- * **Cloud & Platforms:** Azure Databricks, Data Lakehouses, Data Warehouses
- * **Programming Languages:** Python, SQL.
- * **Database Systems:** PostgreSQL, Supabase, Teradata.
- * **ETL & Data Integration:** Informatica PowerCenter,
- * **Data Formats & Storage:** Parquet, Avro, ORC, JSON, CSV
- * **AI & Emerging Technologies:** AI Theory, Large Language Models (LLMs), Prompt Engineering, Model Fine-tuning, NLP, AI Agents, n8n Workflow Automation, Google Colab, Cursor IDE

DATA MODELING & BUSINESS ANALYSIS STRENGTHS

- * **Requirements Translation:** Adept at analyzing raw corporate data and vague business briefs to identify core entities, logical attributes, business keys, and system constraints.
- * **Schema & Relational Architecture:** Highly skilled in constructing scalable relational models and multi-dimensional schemas (Star/Snowflake), aligning technical implementation with high-level enterprise business rules.
- * **Data Integrity & Flow:** Expert understanding of normalization rules, referential integrity, and technical lineage tracking to ensure data reliability across upstream ingestion and downstream analytical consumers.

AI & EMERGING TECHNOLOGY KNOWLEDGE

- * **Comprehensive AI Theory:** Deep foundational understanding of supervised, unsupervised, and reinforcement learning paradigms, neural network configurations, loss optimization, and evaluation metrics.
- * **Enterprise Integration:** Strategic knowledge of leveraging generative AI models, vector embeddings, and semantic search architectures to optimize automated data analysis and corporate decision-making tools.
- * **Workflow Automation:** Proficient in designing agentic workflows using n8n and advanced prompt engineering principles to wrap legacy ETL processes with intelligent metadata monitoring.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | Bangalore, India

Senior Data Engineer (Pittsburgh National Corporation Project) | May 2022 – Present

- * Architected and deployed high-performance Apache Spark data pipelines processing over 50TB of raw information daily, maintaining a 99.9% pipeline uptime across 10,000+ downstream business applications.
- * Led an enterprise-wide CDH-to-CDP migration involving over 500 mainframe-ingested tables, establishing relational entity models that guaranteed zero system downtime and generated \$2M in annual infrastructure savings.
- * Analyzed complex stakeholder business requirements to engineer structural data layouts, utilizing advanced partitioning and bucketing methods to reduce query execution durations by 60% and improve storage efficiency by 40%.
- * Developed comprehensive data quality frameworks leveraging Python and SQL, realizing a 99.8% production data accuracy rate while diminishing data defects by 75%.
- * Designed low-latency real-time data streaming architectures utilizing Apache Kafka and Spark, processing more than 1M events per second with sub-second data availability.
- * Automated enterprise-level batch monitoring configurations across 200+ daily processes using CA7 scheduling tools, reducing manual intervention frameworks by 80%.
- * Optimized big data ecosystem compute performance across Hive, Impala, and Databricks platforms, yielding a 3x enhancement in data processing execution speeds.
- * Mentored a functional delivery squad of 5 junior engineers through regular technical code reviews, technical architecture guidance, and knowledge sharing, boosting team productivity by 50%.

Mapped multi-source enterprise schemas and exported automated anomaly reports into structured Excel PivotTables and dashboards for non-technical business stakeholders to analyze system dependencies

PROJECTS

Enterprise Data Lake Modernization (2023–2024)

- Tech Stack: Azure Databricks, Azure Data Factory, Delta Lake, Python, SQL
- Problem Statement: Legacy distributed infrastructure could not handle analytical performance scaling, leading to high operational costs and lack of transactional ACID compliance.

- Key Responsibilities: Analyzed source system attributes to map out a clean modern lakehouse topology. Implemented logical and physical data layers while configuring Delta Lake storage mechanisms for ACID transaction integrity and historic time-travel analysis.
- Impact / Outcome: Migrated over 100TB of historical data with zero systemic loss, reducing data compute platform overhead costs by 45% through automated multi-tier compression models.

Real-time Analytics Platform (2022–2023)

- Tech Stack: Apache Kafka, Apache Spark Streaming, Machine Learning Models, PostgreSQL
- Problem Statement: Financial transactional verification workflows required sub-second fraud classification to shield downstream account balances from exposure.
- Key Responsibilities: Conducted deep data elements analysis to map streaming JSON payloads to a unified relational model. Integrated statistical machine learning classification modules directly into the low-latency processing loop.
- Impact / Outcome: Built a robust data delivery framework ensuring sub-100ms processing latency with a 35% improvement in fraud identification accuracy while guaranteeing 99.99% infrastructure uptime.

AI-Powered Data Quality Framework (2024)

- Tech Stack: Python, Natural Language Processing (NLP), Machine Learning, n8n, Metadata Repository
- Problem Statement: Manual validation of complex multi-source enterprise schemas introduced delayed delivery times and high data lineage failure rates.
- Key Responsibilities: Developed analytical evaluation processes using natural language processing to read business definitions, automatically identifying entity anomalies, structural shifts, and system dependencies.
- Impact / Outcome: Eliminated manual compliance validation workloads by 90% through end-to-end automated anomaly alerting, significantly reinforcing production data lineage reliability.

EDUCATION

Bachelor of Science in Civil Engineering

Annamacharya Institute of Technology and Sciences | Rajampet, Andhra Pradesh, India

Graduation: 2022

* Relevant Focus Areas: Statistical Inference, Systems Engineering, Complex Data Analysis, Lifecycle Project Management

THOUGHT LEADERSHIP & ENGAGEMENT

- * Technical Author: Active industry writer on Medium specializing in modern big data architecture, scalable pipeline optimization strategies, and next-generation AI integration.
- * Community Engagement: Contributor to open-source distributed computing initiatives, data engineering forums, and internal technical knowledge networks.
- * Speaker: Presenter at internal enterprise tech talks focusing on integrating AI/ML models into distributed core data pipelines.

LANGUAGE

English

Telugu